Principles of Quantum Mechanics

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9th November 2026

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1 Introduction

Recall from IB Quantum Mechanics we have the following:

(i) Wave-particle duality. Waves behave like particles, like light quanta, and vice versa (particles behave like waves).

(ii) We have a wavefunction $\psi(x)$ for a particle which we interpret as a probability density $|\psi(x)|^2$ (i.e. probability is intrinsic to the theory).

(iii) Observables become operators on wavefunctions:

- $\hat{x} \to x$;
- $\hat{p} \to -i\hbar\frac{\partial}{\partial x}$

The lack of commutation, since $[\hat{x}, \hat{p}] = i\hbar$, leads to the uncertainty relation.

(iv) Schrödinger's equation with Hamiltonian H specifies dynamics, and eigenvalues of H are the allowed energy levels of the system.

(v) All of this explains the hydrogen atom structure and the resulting line spectra.

The aims of this course are the following:

(i) Reformulate quantum mechanics in a more powerful, flexible and useful form: the Dirac formulation, which goes beyond wavefunctions. The previous description appears as a particular realisation of our generalisation of quantum states and also leads to alternatives such as

$$\tilde{\psi}(p) \quad \text{with} \quad \hat{x} \to i\hbar\frac{\partial}{\partial p} \quad \text{and} \quad \hat{p} \to p.$$

(ii) This more general approach allows much simpler analysis of known problems (for example, the Simple Harmonic Oscillator), but also lots of new things such as *spin* of particles and symmetries (translations, rotations) and the related conservation laws.

(iii) We will also look at identical particles (bosons and fermions).

(iv) We will develop a framework for quantising more general systems such as EM fields.

(v) We will also set up perturbation theory, which is a practical tool for computation, but we will not heavily consider the detailed applications.

2 Dirac Formalism

2.1 Concepts and Definitions

A quantum system is described at each instant by a *state* $|\psi\rangle$ which belongs to a complex vector space V; any non-zero element of this space V is a possible state. Thus

$$|\psi\rangle, |\phi\rangle \in V \implies \alpha|\psi\rangle + \beta|\phi\rangle \in V \quad \text{for any } \alpha, \beta \in \mathbb{C}.$$

Physically, this is a superposition principle leading to wave-like behaviour, but these states are *not wavefunctions*. There are also *dual* or conjugate states $\langle\psi|$ which belong to the dual space

$V^\dagger$. By definition, states and duals can be combined or paired to give a complex number. The map

$$(\langle\psi|\,, |\phi\rangle) \mapsto \langle\psi|\phi\rangle$$

defines a map from $V^\dagger \times V \to \mathbb{C}$ with

$$\langle\phi|\,(\alpha_1\,|\psi_1\rangle + \alpha_2\,|\psi_2\rangle) = \alpha_1\,\langle\phi|\psi_1\rangle + \alpha_2\,\langle\phi|\psi_2\rangle$$

and

$$(\beta_1\,\langle\phi_1| + \beta_2\,\langle\phi_2|)\,|\psi\rangle = \beta_1\,\langle\phi_1|\psi\rangle + \beta_2\,\langle\phi_2|\psi\rangle\,.$$

The spaces V and $V^\dagger$ come with an inner product which can be described as a one-to-one correspondence between the spaces

$$\begin{aligned} V &\to V^\dagger \\ |\psi\rangle &\to \langle\psi| = (|\psi\rangle)^\dagger\,. \end{aligned}$$

We also have that if we take a linear combination of states to the dual space, the coefficients have to have their complex conjugates taken when mapping from the dual space to V or vice versa. Hence the inner product is

$$|\phi\rangle\,, |\psi\rangle \mapsto \langle\phi|\psi\rangle = (|\phi\rangle)^\dagger\,|\psi\rangle$$

and it is assumed to obey

$$\langle\phi|\psi\rangle = (\langle\psi|\phi\rangle)^*$$

since the inner product is Hermitian, and the inner product defines a positive-definite norm on V:

$$\|\,|\psi\rangle\,\|^2 = \langle\psi|\psi\rangle\,.$$

Remark. Knowing $\langle\phi|\psi\rangle$ for all $\langle\phi|$ determines $|\psi\rangle$ and vice versa.

We can show this by

$$\langle\phi|\psi_1\rangle = \langle\phi|\psi_2\rangle \implies \langle\phi|\chi\rangle = 0,$$

where $|\chi\rangle = |\psi_1\rangle - |\psi_2\rangle$. If this holds for all $\langle\phi|$, we can set $\langle\phi| = \langle\chi|$, so by positive definiteness we have that the norm of $|\chi\rangle$ is zero, and therefore $|\chi\rangle = 0$ too.

We also have that the physical content of any state is unchanged by

$$|\psi\rangle \to \alpha\,|\psi\rangle$$

for any non-zero $\alpha \in \mathbb{C}$. So we will usually normalise to make the state have norm 1, but we can still change it by a phase factor.

Remark. Relative phases can be important in expressions where we're adding a linear combination of elements of V. We also assume the space V is complete so appropriate sequences and series converge in V.

A complete inner product space is called a *Hilbert space*.

Definition. (Operator) An *operator* Q is a linear map on states

$$|\psi\rangle \to Q\,|\psi\rangle\,.$$

The same operator can be thought of as acting to the left on dual states, so

$$\langle\phi| \to \langle\phi|\,Q$$

where the right-hand side is defined by

$$(\langle\phi|\,Q)\,|\psi\rangle = \langle\phi|\,(Q\,|\psi\rangle) = \langle\phi|\,Q\,|\psi\rangle\,.$$

Remark. States and operators $|\psi\rangle, \langle\phi|, Q$ can be dealt with by rulers that mirror those for complex vectors $\mathbf{u}(\text{col}), \mathbf{v}^\dagger(\text{row})$ and a matrix A. So we can compare $\langle\psi|\,Q\,|\phi\rangle$ with $\mathbf{v}^\dagger A\mathbf{u}$ like we do with inner products.

Definition. (Eigenstate) For any operator Q with say that $|\psi\rangle$ non-zero is an *eigenstate* with *eigenvalue* λ if

$$Q\,|\psi\rangle = \lambda\,|\psi\rangle$$

or equivalently

$$\langle\psi|\,Q^\dagger = \lambda^*\,\langle\psi|\,.$$

Remark. For a general Q, λ may be complex.

Definition. (Hermitian Conjugate) For any operator Q, the *Hermitian conjugate* or *adjoint map* $Q^\dagger$ is an operator defined by

$$\langle\psi|\,Q^\dagger = (Q\,|\psi\rangle)^\dagger$$

for any $|\psi\rangle$.

Equivalently

$$\begin{aligned}\langle\phi|\,Q^\dagger\,|\psi\rangle &= (Q\,\langle\phi|)^\dagger\,|\psi\rangle \\ &= \langle\psi|\,Q\,|\phi\rangle^\dagger\end{aligned}$$

for all $|\psi\rangle, |\phi\rangle$.

Definition. (Hermitian) An operator Q is *Hermitian* or *self-adjoint* if

$$Q^\dagger = Q$$

and such operators are called *observables* because they correspond to physical quantities that can be measured, for example position, momentum, energy.

To define a quantum system, we must specify the *Hamiltonian* whose eigenvalues are the allowed energy levels of the system,

$$H|\chi\rangle = E|\chi\rangle.$$

We have the following key results for any Hermitian operator Q.

Proposition. For a Hermitian operator Q,
(i) All eigenvalues are real;
(ii) Eigenstates with distinct eigenvalues are orthogonal;
(iii) Any state can be written as a linear combination of its eigenstates (i.e its matrix representation is diagonalisable).

For a proof of (iii) we have a proof in IB Linear Algebra. We'll cover a proof of (i) and (ii) here.

Proof. Suppose that $Q|\psi\rangle = \lambda|\psi\rangle$. So taking Hermitian conjugates on both sides we have that

$$\begin{aligned}\langle\psi|Q^\dagger &= \lambda^*\langle\psi| \\ &= \langle\psi|Q.\end{aligned}$$

Hence $\langle\psi|Q|\psi\rangle = \lambda\langle\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle$. Since $|\psi\rangle$ is non-zero, it's norm is non-zero hence $\lambda = \lambda^*$.

For the second part let $|n\rangle$ be an eigenstate of Q with eigenvalues $\lambda = q_n$ real. Then we have that

$$\begin{aligned}Q|n\rangle &= q_n|n\rangle \\ Q|m\rangle &= q_m|m\rangle \\ \langle m|Q &= q_m\langle m|\end{aligned}$$

hence $\langle m|Q|n\rangle = q_n\langle m|n\rangle = q_m\langle m|n\rangle$. But we have that $q_m \neq q_n$ so $\langle m|n\rangle = 0$. □

2.2 The Harmonic Oscillator

2.2.1 Construction of the eigenstates

A one-dimensional harmonic oscillator of mass m and frequency ω is a system with position $\hat{x}$, momentum $\hat{p}$ and Hamilitonian

$$H = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2.$$

We seek the energy eigenstates and eigenvalues of H. We do this by introducting *annihilation, creation, and number operators.*

Define

$$a = \left(\frac{m\omega}{2\hbar}\right)^{\frac{1}{2}} \left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)$$
$$a^\dagger = \left(\frac{m\omega}{2\hbar}\right)^{\frac{1}{2}} \left(\hat{x} - \frac{i\hat{p}}{m\omega}\right).$$

Equivalently

$$\hat{x} = \left(\frac{\hbar}{2m\omega}\right)^{\frac{1}{2}} \left(a + a^\dagger\right)$$
$$\hat{p} = \left(\frac{\hbar m\omega}{2}\right)^{\frac{1}{2}} i\left(a^\dagger - a\right).$$

It's easy to check that

$$[\hat{x}, \hat{p}] = i\hbar \iff [a, a^\dagger] = 1.$$

In fact,

$$\begin{aligned} aa^\dagger &= \frac{m\omega}{2\hbar}\hat{x}^2 + \frac{1}{2m\omega\hbar}\hat{p}^2 - \frac{i}{2\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x}) \\ &= \frac{1}{\hbar\omega}H + \frac{1}{2} \end{aligned}$$

but

$$a^\dagger a = \frac{1}{\hbar\omega}H - \frac{1}{2}.$$

This confirms the commutation relation, but we also find that

$$H = \hbar\omega\left(N + \frac{1}{2}\right)$$

with $N = a^\dagger a$, the *number operator* which is clearly Hermitian, so finding eigenvalues and eigenstates of N is equivalent to finding these for H.

Let $|\lambda\rangle$ be any eigenstate of N with eigenvalue λ, so

$$N|\lambda\rangle = \lambda|\lambda\rangle \iff H|\lambda\rangle = E|\lambda\rangle$$

where $E = \hbar\omega\left(\lambda + \frac{1}{2}\right)$. Then assuming that $|\lambda\rangle$ is normalised, write

$$\lambda = \langle\lambda| N |\lambda\rangle = \lambda a^\dagger a|\lambda\rangle = (a|\lambda\rangle)^\dagger (a|\lambda\rangle) = ||a|\lambda\rangle||^2$$

we could have just seen this directly, since N is clearly positive semi-definite from its construction as the outer product of an operator. So we have that $\lambda \geq 0$ and $\lambda = 0 \iff a|\lambda\rangle = 0$.

Now we consider

$$\begin{aligned} N(a^\dagger|\lambda\rangle) &= ([N, a^\dagger] + a^\dagger N)|\lambda\rangle \\ &= (a^\dagger + a^\dagger\lambda)|\lambda\rangle \\ &= (\lambda + 1)(a^\dagger|\lambda\rangle). \end{aligned}$$

Similarly we have that

$$\begin{aligned} N(a\,|\lambda\rangle) &= = ([N,a] + aN)\,|\lambda\rangle \\ &= (-a + a\lambda)\,|\lambda\rangle \\ &= (\lambda - 1)a\,|\lambda\rangle\,. \end{aligned}$$

Here we have used the results for commutators $[N, a^\dagger] = [a^\dagger, a^\dagger] = a^\dagger[a, a^\dagger] = a^\dagger$ and similarily $[N, a] = -a$.

Now the news states with be eigenstates provided their norms are non-zero. We can compute

$$\begin{aligned} ||a\,|\lambda\rangle\,||^2 &= \lambda \\ ||a^\dagger\,|\lambda\rangle\,||^2 &= \langle\lambda|\,aa^\dagger\,|\lambda\rangle = \lambda a^\dagger a + 1\,|\lambda\rangle \\ &= \lambda + 1. \end{aligned}$$

Because of these properties $a^\dagger$ and a are called *creation* (or raising) and *annihilation* (or lowering) operators.

We can't lower below zero, since the eigenvalues are positive so the eigenvalues must be a subset of the positive integers to avoid a contradiction. We can lower and raise all we want, so we can generate any other state with eigenvalues being all of the non-negative integers.

More formally, since we know that $\lambda \geq 0$, so for a given eigenstate $|\lambda\rangle$, if λ is *not* an integer, then

$$a\,|\lambda\rangle\,, a^2\,|\lambda\rangle\,, \ldots, a^n\,|\lambda\rangle$$

are states with eigenvalues

$$\lambda - 1, \lambda - 2, \ldots, \lambda - n$$

but for n sufficiently large $\lambda - n < 0$ which is a contradiction.

By contrast if $\lambda = k \in \mathbb{N}$ then the contradiction is avoided because

$$a\,|k\rangle\,, a^2\,|k\rangle\,, \ldots, a^k\,|k\rangle$$

with eigenvalues

$$k - 1, k - 2, \ldots, 0$$

so we can't lower anymore because our eigenstate has collapsed to zero, i.e. if we force integer eigenvalues the resultant eigenstate is *nilpotent*.

We can use the raising operator $a^\dagger$ to generate the eigenstates with eigenvalues $k + 1, k + 2, \ldots$ forever.

Hence the eigenvalues of N are therefore precisely the non-negative integers and the oscillator has energy levels

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right).$$

From our calculation of norms above, we can choose normalised eigenstates $|n\rangle$ related by

$$\begin{aligned} a^\dagger\,|n\rangle &= \sqrt{n+1}\,|n+1\rangle \\ a\,|n\rangle &= \sqrt{n}\,|n-1\rangle\,. \end{aligned}$$

Starting from any state, $|n\rangle$, we can generate all others. In particular, starting from the *ground state* $|0\rangle$, characterised by

$$a\,|0\rangle = 0$$

we can construct normalised eigenstates

$$|n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n\,|0\rangle\,.$$

Remark. Beware that $|0\rangle$ is *not* zero in V. It is an eigenstate and therefore non-zero by definition.

2.2.2 Importance of the oscillator

(i) A smooth potential $V(x)$ with equilibrium point $x = x_0$ (so that $V'(x_0) = 0$) can be expanded by

$$V(x_0 + x) = V(x_0) + \frac{1}{2}V''(x_0)x^2 + O(x^3)$$

so, if $x = x_0$ is *stable*, then we can set $V''(x_0) = m\omega^2 > 0$ and for displacements x not too large, we can expect the system to be well approximated by a oscillator.

(ii) This can also be applied in the case for a diatomic molecules where we have quantisation of vibrational energies which is important in understanding the internal energy and heat capacities which result in macroscopic consequences.

(iii) More complicated systems can be analysed in terms of normal modes. Coherent motion in which all degrees of freedom oscillate with the same common frequency ω. The general solution for a classical oscillator is

$$x(t) = Ae^{-i\omega t} + A^* e^{i\omega t}$$

where A is a complex constant.

Normal modes for a system with variables x_N are solutions of the form

$$x_N(t) = Au_N e^{-i\omega t} + A^* u_N^* e^{i\omega t}$$

with u_N a complex vector specific to each normal frequency ω.

(iv) For example consider a crystal with a number of atoms $\mathcal{N} \sim 10^{23}$ where the force are approximately elastic and N labels diretions of motion for each atom, so we have $3 \times \mathcal{N}$ values.

(v) Or we consider an EM field, where normal modes are oscillations of electric and mangetic at each point in space are given by

$$\mathbf{E}(\mathbf{x}, t) = A\mathbf{u}(\mathbf{x})e^{-i\omega t} + A^*\mathbf{u}(\mathbf{x})^* e^{i\omega t}.$$

This is exact.

(vi) Normal modes are independent oscillator: treat these as above. For each mode we have

$$a(\omega) \quad \text{and} \quad a(\omega)^\dagger$$

where are ground state $|0\rangle$ with $a(\omega)\,|0\rangle = 0$. Non-ground states created with the raising operators add energy and momentum and behave like particles. In this way we can see that a destroys particles and $a^\dagger$ creates particles.

2.3 Bases and Measurements

2.3.1 Bases of Eigenstates

We know that for any observable Q there is an orthonormal basis of eigenstates $\{|n\rangle\}$ for V,

$$Q\,|n\rangle = q_n\,|n\rangle\,.$$

We have orthonormality $\langle m|n\rangle = \delta_{mn}$ and for any state $|\psi\rangle \in V$ we have that

$$|\psi\rangle = \sum_n \alpha_n\,|n\rangle$$

where $\alpha_n = \langle n|\psi\rangle$. Note that $||\,|\psi\rangle\,||^2 = 1$ if and only if

$$\langle\psi|\psi\rangle = \left(\sum_m \alpha_m^*\,\langle m|\right)\left(\sum_n \alpha_n\,|n\rangle\right) = \sum_n |\alpha_n|^2 = 1.$$

Note that we can write

$$1 = \sum_n |n\rangle\,\langle n|\,.$$

since for any state $|\psi\rangle$ we have that

$$\begin{aligned}|\psi\rangle &= \left(\sum_n |n\rangle\,\langle n|\right)|\psi\rangle \\ &= \sum_n |n\rangle\,(\langle n|\psi\rangle)\end{aligned}$$

In above there may be several independent staets with the same eigenvalue λ. The eigenspace for a given eigenvalue is

$$V_\lambda = \{|\psi\rangle : Q\,|\psi\rangle = \lambda\,|\psi\rangle\}.$$

Note that V_λ has basis $\{|n\rangle : q_n = \lambda\}$. The *degeneracy* of λ is the number of sattes in this basis (i.e the dimension of V_λ). If V_λ has dimension 1, then we say that λ is non-degenerate.

2.3.2 Functions of Operators

It is straight forward to define $f(Q)$ for some operator Q if f is a polynmoial or power series (assuming convergence). For example we can consider $f(Q) = Q^3 + Q$ or $f(Q) = e^Q$ in this way. Important examples are $e^{-iHt/\hbar}$ which is the time-evolution operator and $e^{-ia\hat{p}/\hbar}$ which is the translation operator through some fixed displacement a in the x direction.

More generally, for Q an observable, with basis of eigenstates, we can define

$$f(Q)\,|n\rangle = f(q_n)\,|n\rangle$$

and then defined on any state, provided that $f(q_n)$ makes sense. For example for $f(Q) = \frac{1}{Q}$, we have that

$$\frac{1}{Q}\,|n\rangle = \frac{1}{q_n}\,|n\rangle$$

for $q_n \neq 0$.

In addition we have that

$$Q = \sum_n q_n \left|n\right\rangle \left\langle n\right|$$
$$f(Q) = \sum_n f(q_n) \left|n\right\rangle \left\langle n\right|.$$

Recall our eigenspace V_λ and consider a projection operation P_λ onto this subspace, $V \to V_\lambda$ can be defined by its action on basis states

$$P_\lambda \left|n\right\rangle = \begin{cases} \left|n\right\rangle & q_n = \lambda \\ 0 & \text{otherwise} \end{cases}$$

which can be written as

$$P_\lambda = \sum_{n:\ q_n=\lambda} \left|n\right\rangle \left\langle n\right|.$$

2.3.3 Measurement Axioms

Consider a measurement of Q (observable) when the system is in a normalised state $\left|\psi\right\rangle$. Then

(i) The result in an eigenvalue λ of Q;

(ii) This result is obtained with probability

$$p(\lambda) = ||P_\lambda \left|\psi\right\rangle||^2 = \sum_{n:\ q_n=\lambda} |\alpha_n|^2.$$

(iii) Immediately after the measurement the state is

$$cP_\lambda \left|\psi\right\rangle = c \sum_{n:\ q_n=\lambda} \alpha_n \left|n\right\rangle$$

where c is a constant for normalisation.

It is often the case with λ being a non-generate eigenstate. In that case we just get that $p(\lambda) = |\left\langle n|\psi\right\rangle|^2 = |\alpha_n|^2$ where α_n is often called the amplitude of the state.

We have that

$$\sum_\lambda p(\lambda) = 1.$$

Example. Consider a system with space V with orthonormal basis states

$$\left|1\right\rangle, \left|2\right\rangle, \left|3\right\rangle$$

which are eigenstaets of Q with eigenvalues $q_1 = q_2 = 0$ and $q_3 = 1$. Consider

$$\left|\psi\right\rangle = \frac{1}{\sqrt{6}} \left(2\left|1\right\rangle + \left|2\right\rangle + \left|3\right\rangle\right)$$

So we get that

$$p(1) = \left|\frac{1}{\sqrt{6}}\right|^2 = \frac{1}{6}$$
$$p(0) = \left|\frac{2}{\sqrt{6}}\right|^2 + \left|\frac{1}{\sqrt{6}}\right|^2 = \frac{5}{6}$$

and the state afterwards is $|3\rangle$ if 1 was measured and is

$$\frac{1}{\sqrt{5}}(2\,|1\rangle + |2\rangle)$$

if we measured 0.

Definition. (Expectation value) The *expectation value* or *mean* of Q in a state $|\psi\rangle$ is

$$\langle Q\rangle_\psi = \langle\psi|\, Q\,|\psi\rangle = \sum_\lambda \lambda p(\lambda) = \sum_n q_n |\alpha_n|^2.$$

Definition. (Uncertainty) The *uncertainty* or *variance* of an operator Q in a state $|\psi\rangle$ is

$$(\Delta Q)^2_\psi = \langle (Q - \langle Q\rangle_\psi)^2\rangle_\psi = \langle Q^2\rangle_\psi - \langle Q\rangle^2_\psi.$$

Remark. In thinking about many repeated measurements, we must prepare the system in the same state $|\psi\rangle$ each time.

Remark. If $|\psi\rangle$ is an eigenstate of Q with eigenvalue λ then

$$\langle Q\rangle_\psi = \lambda \quad \text{and} \quad (\Delta Q)_\psi = 0.$$

Quantum mechanical behaviour arises from the fact that certain obserables do not commute. In general, in any state $|\psi\rangle$

$$(\Delta A)_\psi (\Delta B)_\psi \geq \frac{1}{2}\,|\langle [A,B]\rangle_\psi|$$

so $[A,B] \neq 0$ implies that we cannot expect to measure A and B at the same time.

A proof of the general uncertainty relation can be seen from expanding $||(A + i\lambda B)\,|\psi\rangle\,||^2 \geq 0$.

The prototypical case is with position and momentum $\hat{x}, \hat{p}$ which obey

$$[\hat{x}, \hat{p}] = i\hbar$$

in one dimension and $\Delta x \Delta p \geq \frac{\hbar}{2}$.

In three dimensions we have

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

and the uncertainty applies to position and momentum in a particular direction.

2.3.4 Simultaneous measurement and complete commuting sets

If an observable Q does not distinguish states completely because some eigenvalues are degenerate, then we can look for an additional obserable Q' such that Q, Q' are simultaneously measureable (i.e simultaneously diagonalisable) and preserves the eigenspaces of the other operator. This means we have joint eigenstates $|\lambda, \lambda'\rangle$ with

$$Q\,|\lambda,\lambda'\rangle = \lambda\,|\lambda,\lambda'\rangle$$
$$Q'\,|\lambda,\lambda'\rangle = \lambda'\,|\lambda,\lambda'\rangle$$

so $[Q, Q'] = 0$. Conversely if $[Q, Q'] = 0$, it is easy to check that $Q' : V_\lambda \to V_\lambda$ and hence we can find a basis of eigenstates of Q' in V_λ which gives us joint eigenstates for Q and Q'. Since we assume that Q is Hermitian so V is a direct sum of eigenstates, we can do this for all eigenstates, giving us a complete basis of joint eigenstates, which we call $|\lambda, \lambda'\rangle$.

Definition. (Complete commuting set) A set of observables $\{Q, Q', Q'', \dots\}$ is a *complete commuting set* if their pairwise commute. This implies that there is a basis of joint eigenstates, and the set of eigenvalues $\lambda, \lambda', \lambda'', \dots$ label states uniquely (up to normalisation)

Remark. Equivalently if A is any observable such that $[A, Q] = [A, Q'] = \cdots = 0$ then $A = f(Q, Q', \dots)$.

Example. Consider the previous example with orthonormal satets $|1\rangle, |2\rangle, |3\rangle$ and eigenstates of Q with $\lambda = 0$ for $|1\rangle$ and $|2\rangle$ and $\lambda = 1$ for $|3\rangle$. Now consider a second observabke Q' with $Q'|1\rangle = |2\rangle, Q'|2\rangle = |1\rangle, Q'|3\rangle = 0$. Now by writing a general vector in the orthonormal basis, we have that $[Q, Q'] = 0$. Now let

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|1\rangle \pm |2\rangle)$$

so then

$$Q'|\pm\rangle = \pm|\pm\rangle$$

Q, Q' is a complete commuting set with basis of joint eigenstates $|+\rangle, |-\rangle, |3\rangle$.

2.4 Bases, Representations and Wavefunctions

2.4.1 Discrete and Continuous Bases

Consider orthonormal basis $\{|n\rangle\}$ with n discrete label. We can regard all states and operators as row or column vectors and matrices respectively

$$|\psi\rangle = \sum_n \alpha_n |n\rangle \iff \alpha_n \qquad = \langle n|\psi\rangle$$
$$|\phi\rangle = \sum_n \beta_n |n\rangle \iff \beta_n \qquad = \langle n|\phi\rangle$$

and inner product

$$\langle\phi|\psi\rangle = \sum_n \beta_n^* \alpha_n.$$

For an operator A,

$$A|n\rangle = \sum_m |m\rangle A_{mn}$$

where A_{mn} is a complex matrix with entries $\langle m| A |n\rangle$.

It's simple to check that

$$|\phi\rangle = A|\psi\rangle \iff \beta_m = \sum_n A_{mn}\alpha_n$$

and also if B is another operator with matrix $B_{mn} = \langle m| B |n\rangle$ then

$$(AB)_{mn} = \sum_p A_{mp} B_{pn}$$

i.e matrix multiplication.

The choice of basis provides a representation of the quantum states and operators. We can also allow the index n to be continuous. This requires modifications such as the following:

$$\begin{aligned} \sum_n &\longrightarrow \int \mathrm{d}n \\ \langle n|m\rangle = \delta_{nm} &\longrightarrow \delta(n-m) \\ |\psi\rangle = \sum_n \alpha \, |n\rangle &\longrightarrow \int \mathrm{d}n \; \alpha_n \, |n\rangle \end{aligned}$$

and the completeness relation goes to

$$I = \sum_n |n\rangle \langle n| \longrightarrow \int \mathrm{d}n \, |n\rangle \langle n| \, .$$

Note that if basis a bove arise as eigenstates of an obserable then

$$|\alpha_n|^2 = |\langle n|\psi\rangle|^2 \longrightarrow \text{probability density for continuous range } n.$$

2.4.2 Position and Momentum Bases

Considera particle in one dimension with position $\hat{x}$ and momentum $|p\rangle$ obeying the relation $[\hat{x}, \hat{p}] = i\hbar$. Let $|x\rangle$ be the position eigenstates

$$\hat{x} \, |x\rangle = x \, |x\rangle$$

which are continuous eigenvalues with

$$\langle x|x'\rangle = \delta(x - x'), \quad \int \mathrm{d}x \, |x\rangle \langle x| = I.$$

Any state can be written as

$$|\psi\rangle = \int \mathrm{d}x \; \psi(x) \, |x\rangle$$

with $\psi(x) = \langle x|\psi\rangle$ and this is the usual position space wavefunction. From our remarks in the previous section $|\psi(x)|^2$ is a probability density for measuring position.

Our inner product of two wavefunctions from Part IB Quantum mechanics is recovered through

$$\begin{aligned} \langle \phi|\psi\rangle &= \left(\int \mathrm{d}x \; \phi(x)^* \, \langle x| \right) \left(\int \mathrm{d}x' \; \psi(x') \, |x\rangle' \right) \\ &= \int \mathrm{d}x \int \mathrm{d}x' \; \phi(x)^* \psi(x') \, \langle x|x'\rangle = \int \mathrm{d}x \; \phi(x)^* \psi(x) \end{aligned}$$

and

$$|| \, |\psi\rangle \, ||^2 = \langle\psi|\psi\rangle = \int \mathrm{d}x |\psi(x)|^2 = 1$$

if normalised.

Similarly, define momentum eigenstate $|p\rangle$ with

$$\hat{p} \, |p\rangle = p \, |p\rangle$$

and

$$\langle p|p'\rangle = \delta(p - p'), \quad \int \mathrm{d}p \, |p\rangle \, \langle p| = I.$$

Now we can expand

$$|\psi\rangle = \int \mathrm{d}p \; \tilde{\psi}(p) \, |p\rangle$$

with $\tilde{\psi}(p) = \langle p|\psi\rangle$ momentum space wavefunction and $|\tilde{\psi}(p)|^2$ probability distribution for measurements of momentum.

To proceed, we need the following result that

$$\langle x|p\rangle = \frac{1}{(2\pi\hbar)^{1/2}} e^{ipx/\hbar}$$
$$\langle p|x\rangle = \frac{1}{(2\pi\hbar)^{1/2}} e^{-ipx/\hbar}$$

which we will assume holds. Consider the action of $\hat{x}$ and $\hat{p}$ on position space wavefunctions. So suppose we have $\hat{x} \, |\psi\rangle$ has wavefunction

$$\langle x| \, \hat{x} \, |\psi\rangle = x \, \langle x|\psi\rangle = x\psi(x).$$

Similarly if we have $\hat{p} \, |\psi\rangle$, the wavefunction is

$$\begin{aligned} \langle x| \, \hat{p} \, |\psi\rangle &= \int \mathrm{d}p \, \langle x| \, \hat{p} \, |p\rangle \, \langle p|\psi\rangle \\ &= \int \mathrm{d}p \; p \, \langle x|p\rangle \, \langle p|\psi\rangle \end{aligned}$$

and note that $p \, \langle x|p\rangle = -i\hbar \frac{\partial}{\partial x} \, \langle x|p\rangle$, so hence

$$\begin{aligned} \langle x| \, \hat{p} \, |\psi\rangle &= -i\hbar \frac{\partial}{\partial x} \int \mathrm{d}p \;\; \langle x|p\rangle \, \langle p|\psi\rangle \\ &= -i\hbar \frac{\partial}{\partial x} \, \langle x|\psi\rangle \\ &= -i\hbar\psi'(x). \end{aligned}$$

which gives us the action of $\hat{x}$ and $\hat{p}$ on position space wavefunctions. Similarly on momentum space wavefunctions we calculate $\langle p| \, \hat{x} \, |\psi\rangle$ and $\langle p| \, \hat{p} \, |\psi\rangle$ and the result is that

$$\hat{x} \longrightarrow i\hbar \frac{\partial}{\partial p}$$
$$\hat{p} \longrightarrow p.$$

The relationship between $\psi(x)$ and $\tilde{\psi}(p)$ also follows from our assumed result, as

$$\begin{aligned}\tilde{\psi}(p) = \langle p|\psi\rangle &= \int \mathrm{d}x \, \langle p|x\rangle \, \langle x|\psi\rangle \\ &= \frac{1}{(2\pi\hbar)^{1/2}} \int \mathrm{d}x \; e^{-ipx/\hbar}\psi(x)\end{aligned}$$

and

$$\begin{aligned}\psi(x) = \langle x|\psi\rangle &= \frac{1}{(2\pi\hbar)^{1/2}} \int \mathrm{d}p \, \langle x|p\rangle \, \langle p|\psi\rangle \\ &= \frac{1}{(2\pi\hbar)^{1/2}} \int \mathrm{d}p \; e^{ixp/\hbar}\tilde{\psi}(p).\end{aligned}$$

Now given $H(\hat{x}, \hat{p})$ we can consider the time independent Schrödinger equation

$$H\,|\psi\rangle = E\,|\psi\rangle\,.$$

If $H = \frac{\hat{p}^2}{2m} + V(\hat{x})$ then on $\psi(x)$ we get that

$$H \to -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

and on $\tilde{\psi}(p)$ we get that

$$H \to \frac{p^2}{2m} + V\left(i\hbar\frac{\partial}{\partial p}\right).$$

It may be easy to interpret the potential term in momentum space for something like $V(x) = \lambda x^n$ which gives

$$V\left(i\hbar\frac{\partial}{\partial p}\right) = \lambda(i\hbar)^n\left(\frac{\partial}{\partial p}\right)^n$$

for a positive integer n. More generally this isn't obvious what the potential term in momentum space means. We will argue from first principles as follows.

Consider

$$\begin{aligned}\langle p|\,V(\hat{x})\,|\psi\rangle &= \int \mathrm{d}x \, \langle p|\,V(\hat{x})\,|x\rangle \, \langle x|\psi\rangle \\ &= \int \mathrm{d}x \int \mathrm{d}p' \, \langle p|\,V(\hat{x})\,|x\rangle \, \langle x|p'\rangle \, \langle p'|\psi\rangle \\ &= \int \mathrm{d}p' \left(\int \mathrm{d}x \; V(x) \, \langle p|x\rangle \, \langle x|p'\rangle \, \langle p'|\psi\rangle\right) \\ &= \int \mathrm{d}p' \int \mathrm{d}x \; V(x)\frac{e^{-i(p-p')x/\hbar}}{2\pi\hbar}\tilde{\psi}(p) \\ &= \frac{1}{(2\pi\hbar)^{1/2}} \int \mathrm{d}p' \; \tilde{V}(p-p')\tilde{\psi}(p')\end{aligned}$$

Which is a reproduction of the convolution theorem. So $H\,|\psi\rangle = E\,|\psi\rangle$ becomes

$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x)$$

in position form. Or

$$\frac{p^2}{2m}\tilde{\psi}(p) + \frac{1}{(2\pi\hbar)^{1/2}}\int \mathrm{d}p'\tilde{V}(p-p')\tilde{\psi}(p') = E\tilde{\psi}(p)$$

in momentum space.

To generalise from one to three dimensions we use the commutation relation that

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}.$$

One complete commuting set is

$$\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2, \hat{x}_3)$$

with joint eigenstates

$$\hat{x}_i \left|\mathbf{x}\right\rangle = x_i \left|\mathbf{x}\right\rangle$$

which are orthonormal, so $\langle \mathbf{x}|\mathbf{x}'\rangle = \delta^{(3)}(\mathbf{x}-\mathbf{x}')$.

An alternative set

$$\hat{\mathbf{p}} = (\hat{p}_1, \hat{p}_2, \hat{p}_3)$$

with joint eigenstates

$$\hat{p}_i \left|\mathbf{p}\right\rangle = p_i \left|\mathbf{p}\right\rangle$$

which are orthonormal in the same sense as the position operator. For any state $|\psi\rangle$, we can expand

$$\left|\psi\right\rangle = \int \mathrm{d}^3\mathbf{x}\ \psi(\mathbf{x})\left|\mathbf{x}\right\rangle = \int \mathrm{d}^3\mathbf{p}\ \tilde{\psi}(\mathbf{p})\left|\mathbf{p}\right\rangle$$

where $\psi(\mathbf{x}) = \langle \mathbf{x}|\psi\rangle$ and $\tilde{\psi}(\mathbf{p}) = \langle \mathbf{p}|\psi\rangle$. We also get that

$$\langle \mathbf{x}|\mathbf{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$$

and the wavefunctions are related by

$$\tilde{\psi}(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}}\int \mathrm{d}^3\mathbf{x}\ e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\psi(\mathbf{x})$$

and the corresponding Fourier inverse.

2.4.3 Position and Momentum Eigenstates

Now we return to one dimension and seek to justify the result for $\langle x|p\rangle$ using only the commutator relation. Given $|x\rangle$ with $\hat{x}\left|x\right\rangle = x\left|x\right\rangle$, consider

$$T(a) = e^{-ia\hat{p}/\hbar} = \sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{-ia}{\hbar}\right)^n \hat{p}^n$$

and note that

$$[\hat{x}, T(a)] = \left[\hat{x}, \frac{-ia\hat{p}}{\hbar}\right] T(a) = aT(a).$$

Hence

$$\begin{aligned}\hat{x}(T(x)\,|x\rangle) &= ([\hat{x}, T(a)] + T(a)\hat{x})\,|x\rangle \\ &= (x+a)(T(x)\,|x\rangle)\end{aligned}$$

so we can think of $T(a)\,|x\rangle$ as a translated eigenstate.

Similarly consider $S(b) = e^{ib\hat{x}/\hbar}$ and note that $[\hat{p}, S(b)] = bS(b)$. Hence

$$\hat{p}\,|p\rangle = p\,|p\rangle \implies \hat{p}(S(b)\,|p\rangle) = (p+b)(S(b)\,|p\rangle)$$

so $S(b)$ gives a translation in momentum space.

Now take fixed values x_0 and p_0 and define all eigenstates by

$$|x_0 + a\rangle = T(a)\,|x_0\rangle\,, \quad |p_0 + b\rangle = S(b)\,|p_0\rangle\,.$$

Now observe that

$$\begin{aligned}\langle x_0 + a | p_0 + b\rangle &= \langle x_0|\, e^{ia\hat{p}/\hbar}\,|p_0 + b\rangle \\ &= \langle x_0|\, e^{ia(p_0+b)/\hbar} e^{ib\hat{x}/\hbar}\,|p_0\rangle \\ &= e^{(ab+ap_0+bx_0)/\hbar}\,\langle x_0|p_0\rangle\,.\end{aligned}$$

Now choose $\langle x_0|p_0\rangle = \frac{1}{\sqrt{2\pi\hbar}}$ for correct and consistent normalisation. Then set $x_0 = p_0 = 0$, so we get

$$\langle a|b\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{iab/\hbar}.$$

3 Evolution in Time and Pictures

3.1 The Schrödinger Equation and the Schrödinger Picture

The evolution in time of state $|\psi(t)\rangle$ is governed (in absence of external influences of measurement) by the Schödinger equation

$$i\hbar \frac{\partial}{\partial t}\,|\psi(t)\rangle = H\,|\psi(t)\rangle$$

where $H = H^\dagger$ is the Hamiltonian.

Equivalently

$$-i\hbar \frac{\partial}{\partial t}\,\langle\psi(t)| = \langle\psi(t)|\,H.$$

This implies that

$$\frac{\partial}{\partial t}\,(\langle\psi(t)|\psi(t)\rangle) = 0$$

so that the normalisation of $|\psi(t)\rangle$ is preserved in time.

The solution of the Schrödinger equation can be written as

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle$$

where $U(t) = e^{-iHt/\hbar}$ is the time-evolution operator, assuming that H is independent of t. We can check immediately from $i\hbar\frac{\partial U(t)}{\partial t} = HU(t)$.

Recall that time-evolution problems can be conveniently solved by working with basis of energy eigenstates $\{|n\rangle\}$ with

$$H|n\rangle = E_n|n\rangle .$$

with

$$e^{-iE_nt/\hbar}|n\rangle$$

being a solution of the time-dependent Schrödinger equation.

Generally if

$$|\psi(0)\rangle = \sum_n \alpha_n |n\rangle$$

where

$$H|n\rangle = E_n|n\rangle$$

then

$$|\psi(t)\rangle = \sum_n \alpha_n e^{-iE_nt/\hbar}|n\rangle .$$

Example. Consider a system with two energy eigenstates $|1\rangle$ and $|2\rangle$. Consider measuring Q defined by

$$Q|1\rangle = |2\rangle , \quad Q|2\rangle = |1\rangle$$

So $[Q, H] \neq 0$. The eigenstates of Q are

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|1\rangle \pm |2\rangle)$$

with eigenvalues $q_\pm = \pm 1$. Take the initial state

$$|\psi(0)\rangle = |+\rangle .$$

This implies that

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-iE_1t/\hbar}|1\rangle + e^{-iE_2t/\hbar}|2\rangle\right)$$

so the probability of measuring Q at time t and getting ± 1 is

$$\begin{aligned} |\langle\pm|\psi(t)\rangle|^2 &= \left|\frac{1}{2}(\langle 1| \pm |2\rangle)(e^{-iE_1t/\hbar}|1\rangle + e^{-iE_2t/\hbar}|2\rangle)\right|^2 \\ &= \left|\frac{1}{2}(e^{-iE_1t/\hbar} \pm e^{-iE_2t/\hbar})\right|^2 \\ &= \begin{cases} \cos^2(E_1 - E_2)t/2\hbar \\ \sin^2(E_1 - E_2)t/2\hbar \end{cases} . \end{aligned}$$

In the basis

$$|1\rangle \longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$|2\rangle \longrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

then

$$H \longrightarrow \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}$$

and

$$U = e^{-iHt/\hbar} \longrightarrow \begin{pmatrix} e^{-iE_1 t/\hbar} & 0 \\ 0 & e^{-iE_2 t/\hbar}. \end{pmatrix}$$

But instead we can work in the alternate basis where

$$|+\rangle \longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$|-\rangle \longrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

So we have that

$$\begin{aligned} H\,|\pm\rangle &= \frac{1}{\sqrt{2}}(E_1\,|1\rangle + E_2\,|2\rangle) \\ \langle\pm|\,H\,|\pm\rangle &= \frac{1}{2}(E_1 + E_2) = \varepsilon \\ \langle\mp|\,H\,|\mp\rangle &= \frac{1}{2}(E_1 - E_2) = \hbar\lambda \end{aligned}$$

where we define ε and λ to be constants. Then we have that

$$H \longrightarrow \begin{pmatrix} \varepsilon & \hbar\lambda \\ \hbar\lambda & \varepsilon \end{pmatrix} = \varepsilon I + \hbar\lambda M$$

where I is the identity matrix and M is the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ so now we have that

$$\begin{aligned} U(t) = e^{-iHt/\hbar} &= e^{-it(\varepsilon I + \lambda\hbar M)/\hbar} \\ &= e^{-it\varepsilon/\hbar} e^{-it\lambda M}. \end{aligned}$$

Note that $M^2 = I$ and hence we have that

$$\begin{aligned} \exp(-it\lambda M) &= \sum_{r=0}^{\infty} \frac{1}{(2r)!}(-i\lambda r)^{2r} I + \sum_{r=0}^{\infty} \frac{1}{(2r+1)!}(-i\lambda t)^{2r+1} M \\ &= (\cos\lambda t)I - i(\sin\lambda t)M. \end{aligned}$$

So we have that

$$|\psi(0)\rangle = |+\rangle \longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

so

$$|\psi(t)\rangle = e^{-iHt/\hbar}\,|+\rangle \longrightarrow e^{-it\varepsilon/\hbar} \begin{pmatrix} \cos\lambda t \\ -i\sin\lambda t \end{pmatrix}$$

which is therefore

$$e^{-it\varepsilon/\hbar}(\cos\lambda t\,|+\rangle - i\sin\lambda t\,|-\rangle).$$

Now compare with previous result for probability of measuring W and finding ± 1.

3.2 Heisenberg Picture and Heisenberg Equations

3.2.1 Unitary operators

Physical predictions in quantum mechanics are given by probability amplitudes, eigenvalues, expectation values, etc, hence we have expressions like

$$\langle\psi|\phi\rangle\,, \quad \langle\phi|\,A\,|\psi\rangle$$

and similar things.

Definition. (Unitary) An operator U is called *unitary* if

$$UU^\dagger = U^\dagger U = I$$

or equivalently

$$U^\dagger = U^{-1}.$$

Given any such U consider a map on states and operators taking

$$\begin{aligned} |\psi\rangle &\longrightarrow U\,|\psi\rangle = |\psi'\rangle \\ \langle\psi| &\longrightarrow \langle\psi|\,U^\dagger = \langle\psi'| \\ A &\longrightarrow UAU^\dagger = A'. \end{aligned}$$

Under this, all physical predictions remain unchanged sinc we have that

$$\langle\phi|\psi\rangle \longrightarrow \langle\phi'|\psi'\rangle = \langle\phi|\,U^\dagger U\,|\psi\rangle = \langle\phi|\psi\rangle$$

and

$$\langle\phi|\,A\,|\psi\rangle \longrightarrow \langle\phi'|\,A'\,|\psi'\rangle = \langle\phi|\,U^\dagger U A U^\dagger U\,|\psi\rangle = \langle\phi|\,A\,|\psi\rangle\,.$$

Furthermore we have that

$$C = AB \longrightarrow C' = A'B'$$

and

$$C = [A, B] \longrightarrow C' = [A', B'].$$

We also have that if Q is Hermitian, then Q' is also Hermitian. We also have the implication that

$$Q\,|\psi\rangle = \lambda\,|\psi\rangle \implies Q'\,|\psi'\rangle = \lambda\,|\psi'\rangle$$

with the same eigenvalue λ.

3.2.2 Changing Pictures

In the Schrödinger picture we have that

$$\begin{aligned} &i\hbar\frac{\partial}{\partial t}\,|\psi(t)\rangle = H\,|\psi(t)\rangle \\ \implies &|\psi(t)\rangle = U(t)\,|\psi(0)\rangle \end{aligned}$$

where $U(t) = \exp(-iHt/\hbar)$ is unitary. In fact we have that $U(t)^\dagger = U(-t) = U(t)^{-1}$ and that $U(t_1)U(t_2) = U(t_1 + t_2)$. The states depend on t but observables do not in the Schrödinger

picture. To pass to the Heisenberg picture we shift t-dependence from states to operators as follows. In the Schrödinger picture we have states and operators

$$|\psi(t)\rangle_{\mathrm{S}} \quad A_{\mathrm{S}}.$$

But in the Heisenberg picture we have

$$\begin{aligned} |\psi\rangle_{\mathrm{H}} &= U^{\dagger} |\psi(t)\rangle_{\mathrm{S}} = e^{itH/\hbar} |\psi(t)\rangle_{\mathrm{S}} = |\psi(0)\rangle_{\mathrm{S}} \\ A_{\mathrm{H}}(t) &= U^{\dagger} A_{\mathrm{S}} U = e^{itH/\hbar} A_{\mathrm{S}} e^{-itH/\hbar} \end{aligned}$$

and because the transformation $U(t)$ is unitary we have that

$$\langle \phi(t)| A_{\mathrm{S}} |\psi(t)\rangle_{\mathrm{S}} = \langle \phi| A_{\mathrm{H}}(t) |\psi\rangle_{\mathrm{H}}$$

Note that $H_{\mathrm{H}} = H_{\mathrm{S}} = H$.

To specify dynamics in the Heisenberg picture consider

$$\begin{aligned} \frac{d}{dt} A_{\mathrm{H}}(t) &= \frac{d}{dt} \left(e^{iHt/\hbar} A_{\mathrm{S}} e^{-iHt/\hbar} \right) \\ &= \frac{iH}{\hbar} e^{iHt/\hbar} A_{\mathrm{S}} e^{-iHt/\hbar} - e^{iHt/\hbar} A_{\mathrm{S}} e^{-iHt/\hbar} \frac{iH}{\hbar} \\ &= \frac{i}{\hbar} [H, A_{\mathrm{H}}(t)] \end{aligned}$$

or

$$i\hbar \frac{d}{dt} A_{\mathrm{H}}(t) = [A_{\mathrm{H}}(t), H]$$

which is the Heisenberg equations of motion.

Example. Consider a particle in one dimension $\hat{x}(t), \hat{p}(t)$ in the Heisenberg picture. Consider $H = \frac{\hat{p}^2}{2m} + V(\hat{x})$ so the Heisenberg equations of motion become

$$\frac{d}{dt} \hat{x}(t) = \frac{1}{i\hbar} [\hat{x}(t), H] = \frac{1}{m} \hat{p}(t)$$

and

$$\frac{d}{dt} \hat{p}(t) = \frac{1}{i\hbar} [\hat{p}(t), H] = -V'(\hat{x}(t))$$

using $[\hat{x}(t), \hat{p}(t)] = i\hbar$.

Taking expectation values in any state $|\psi\rangle$ (now independent of t in the Heisenberg picture) gives

$$\frac{d}{dt} \langle \hat{x} \rangle = \frac{1}{m} \langle \hat{p} \rangle$$

and

$$\frac{d}{dt} \langle \hat{p} \rangle = -\langle V'(\hat{x}) \rangle$$

which is Ehrenfest's theorem, which is true in all pictures. For some potentials we can solve the Heisenberg question. For example if $V = 0$ we have a free particle, so

$$\frac{d}{dt} \hat{p}(t) = 0$$

which means that $\hat{p}(t) = \hat{p}(0)$ which is a constant operator. And therefore $\hat{x}(t) = \hat{x}(0) + \frac{\hat{p}(0)}{m}t$ like classical dynamics. If we have $V = \frac{1}{2}m\omega^2 x^2$ for a oscillator then we get the set of equations

$$\begin{cases} \frac{d}{dt}\hat{x}(t) = \frac{1}{m}\hat{p}(t) \\ \frac{d}{dt}\hat{p}(t) = -m\omega^2\hat{x}(t) \end{cases}$$

which gives that

$$\frac{d^2}{dt^2}\hat{x} + \omega^2\hat{x} = 0$$
$$\frac{d^2}{dt^2}\hat{p} + \omega^2\hat{p} = 0$$

which as a solution gives

$$\begin{cases} \hat{x}(t) = \hat{x}(0)\cos\omega t + \frac{\hat{p}(0)}{m\omega}\sin\omega t \\ \hat{p}(t) = \hat{p}(0)\cos\omega t - m\omega\hat{x}(0)\sin\omega t \end{cases}$$

or equivalently

$$\begin{cases} \hat{x}(t) = \left(\frac{\hbar}{2m\omega}\right)^{1/2}(ae^{-i\omega t} + a^\dagger e^{i\omega t}) \\ \hat{p}(t) = \left(\frac{\hbar m\omega}{2}\right)^{1/2}\frac{1}{i}(ae^{-i\omega t} - a^\dagger e^{i\omega t}) \end{cases}$$

3.3 Canonical Quantisation (Non-examinable)

This is the final step in Dirac's approach to constructing quantum mechanics. How do we pass from a general classical system to its quantum version? Any classical system can be described by a set of generalised positions $x_i(t)$ and momenta $p_i(t)$ with $1 \leq i \leq N$ which may include angles and angular momenta. These are coordinates in *phase space*. Dynamics are governed by some Hamiltonian $H(x_i, p_i)$. The *Poisson bracket* of any functions $f(x_i, p_i), g(x_i, p_i)$ is defined by

$$\{f, g\} = \sum_i \frac{\partial f}{\partial x_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial x_i}$$

which is a new function on phase space. The properties of Poisson brackets closely resemble those of commutators. Dynamics in this formulation given by Hamilton's equation

$$\frac{df}{dt} = \{f, H\}.$$

In *cononical quantisation* define quantum theory by taking classical funtions to quantum operators and Poisson brackets determine commutators.

$$[\hat{f}, \hat{g}] = i\hbar\widehat{\{f, g\}}$$

which recovers $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$ and Heisenberg's equation from Hamilton's equation.

4 Composite Systems and Identical Particles

4.1 Tensor Products

This is a general way of constructing quantum systems from simplier subsystems. Suppose that

$$|\psi\rangle \in V_1, \quad |\phi\rangle \in V_2$$

are states in spaces for two systems. The tensor product space

$$V = V_1 \otimes V_2$$

which consists of all linear combinations of tensor product states

$$|\psi\rangle \otimes |\phi\rangle$$

subject to

(i) $(|\psi\rangle + |\psi'\rangle) \otimes |\phi\rangle = |\psi\rangle \otimes |\phi\rangle + |\psi'\rangle \otimes |\phi\rangle$;

(ii) $|\psi\rangle \otimes (|\phi\rangle + |\phi'\rangle) = |\psi\rangle \otimes |\phi\rangle + |\psi\rangle \otimes |\phi'\rangle$;

(iii) $(\alpha\,|\psi\rangle) \otimes |\phi\rangle = |\psi\rangle \otimes (\alpha\,|\phi\rangle) = \alpha(|\psi\rangle \otimes |\phi\rangle)$.

and similarly for duals.

The inner product on V satisfies

$$(\langle\psi'| \otimes \langle\phi'|)(|\psi\rangle \otimes |\phi\rangle) = \langle\psi'|\psi\rangle\,\langle\phi'|\phi\rangle$$

and extend by linearity. Given operators A and B on V_1 and V_2 respectively, we define an operator $A \otimes B$ on V by

$$(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = (A\,|\psi\rangle) \otimes (B\,|\phi\rangle)$$

and again extend by linearity. In particular, $A \longrightarrow A \otimes I$ and $B \longrightarrow I \otimes B$. Operators of this form commute for any A, B. Consider a particle in two dimensions with position operators $\hat{x}_1, \hat{x}_2$, The basis of join eigensates constructed as $|x_1, x_2\rangle$ we can take $|x_1, x_2\rangle = |x_1\rangle \otimes |x_2\rangle$. This makes it obvious that $[\hat{x}_i, \hat{x}_j] = 0$ for $i \neq j$.

The wavefunction for $|\psi\rangle \otimes |\phi\rangle$ is now given by

$$\begin{aligned}\langle x_1, x_2|\,(|\psi\rangle \otimes |\phi\rangle) &= (\langle x_1| \otimes \langle x_2|)(|\psi\rangle \otimes |\phi\rangle) \\ &= \langle x_1|\psi\rangle\,\langle x_2|\phi\rangle \\ &= \psi(x_1)\phi(x_2).\end{aligned}$$

This gives a correspondence between tensor product state and separable wavefunctions. Hence there is correspondence between the tensor product space and all linaer combinations of such wavefunctions.

Remark. If we have basis $\{|n\rangle\}$ for V_1 and $\{|m\rangle\}$ for V_2 then $\{|n\rangle \otimes |m\rangle\}$ is a basis for V. If basis are finite then the dimension of the tensor produce space is the product of the dimensions.

Example. Consider the two dimensional isotropic oscillator where isotropic means we have the same angular velocity in all directions. So

$$H = \frac{1}{2m}(\hat{p}_1^2 + \hat{p}_2^2) + \frac{1}{2}m\omega^2(\hat{x}_1^2 + \hat{x}_2^2) = H_1 + H_2$$

where

$$H_i = \frac{1}{2m}\hat{p}_i^2 + \frac{1}{2}m\omega^2\hat{x}_i^2 = \hbar\omega\left(N_i + \frac{1}{2}\right)$$

where $N_i = a_i^\dagger a_i$ with $[a_i, a_j^\dagger] = \delta_{ij}$. Note that $[N_1, N_2] = 0$ so we can construct simulataneous eigenstates like

$$|n_1, n_2\rangle = |n_1\rangle \otimes |n_2\rangle$$

where

$$H\left|n_1,n_2\right\rangle = \left(H_1\left|n_1\right\rangle\right)\otimes\left|n_2\right\rangle + \left|n_1\right\rangle\otimes\left(H_2\left|n_2\right\rangle\right) = E_{n_1n_2}\left|n_1,n_2\right\rangle$$

with $E_{n_1,n_2} = \hbar\omega(n_1+n_2+1)$ so our eigenstates as wavefunctions are

$$\psi_{n_1n_2}(x_1,x_2) = \langle x_1,x_2|n_1,n_2\rangle = \psi_{n_1}(x_1)\psi_{n_2}(x_2)$$

4.2 Spin

Experiements show that particles carry an *internal* degree of freedom called *spin* or intrinsic angular momentum. Even if the particle appears to be "elementary" or "point-like" its space of states is of the form

$$V = V_{\text{space}}\otimes V_{\text{spin}}$$

i.e that position operators are not (in general) a complete commuting set by themselves. There are other observables Q acting just on V_{spin} with $[\hat{x}_i, Q] = 0$. We will return to these operators later, for now we'll just focus on the states.

Each kind of particle ha a definite total spin s which is a half integer

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

which is a part of the particle's nature like mass or charge. For spin s , a particle of non-zero mass has $2s+1$ basis states in V_{spin} labelled by $r = s, s-1, \dots, -s$. We may sometimes use $\left|\uparrow\right\rangle, \left|\downarrow\right\rangle$ to correspond to $\left|\frac{1}{2}\right\rangle$ and $\left|-\frac{1}{2}\right\rangle$.

The existance of spin states is revealed by experiements like the Stern-Gerlach experiment where we can separate a beam of particles by spin by using an inhomogeneous electromagnetic field which causes a difference effect (force) on different spin states.

A general state $|\psi\rangle$ in V has a $(2s+1)$-component wavefunction

$$\psi(\mathbf{x},r) = \langle \mathbf{x},r|\psi\rangle$$

where $|\psi(\mathbf{x},r)|^2$ is the probability density for position of a particle with spin state r. A mixed/hybrid notation is also common with

$$\sum_r \psi(\mathbf{x},r)\left|r\right\rangle$$

4.3 Bosons and Fermions

Consider particles labelled by $a = 1, 2, \dots, N$ and let V_a be a space of states for each particle with a basis $\{|\mathbf{x}_a, r_a\rangle\}$. The general multiparticle states belong to

$$\bigotimes_{i=1}^{N} V_i$$

with basis states

$$\left|\mathbf{x}_1,r_1,\mathbf{x}_2,r_2,\dots,\mathbf{x}_N,r_N\right\rangle = \bigotimes_{i=1}^{N}\left|\mathbf{x}_i,r_i\right\rangle.$$

4.3.1 Bose-Einstein and Fermi-Dirac Statistics

If particles above are identical then $V_a \cong V$. Consider the simple case with $N = 2$. Define an operator U which exchanges particles by actin on basis states

$$U\left|\mathbf{x}_1, r_1, \mathbf{x}_2, r_2\right\rangle = \left|\mathbf{x}_2, r_2, \mathbf{x}_1, r_1\right\rangle$$

so the action on a general two particle state is

$$U\left|\Psi\right\rangle = \eta\left|\Psi\right\rangle$$

since $\left|\Psi\right\rangle$ and $U\left|\Psi\right\rangle$ are physically equivalent states. But $U^2 = 1$ forms an action on basis states so $\eta^2 = 1$ hence η is a second root of unity i.e ± 1. Thus 2-particle states do not belong merely to $V \otimes V$ spanned by $\left|\psi\right\rangle \otimes \left|\phi\right\rangle$ but either to $(V \otimes V)_{\mathrm{S}}$ spanned by $\left|\psi\right\rangle \otimes \left|\phi\right\rangle + \left|\phi\right\rangle \otimes \left|\psi\right\rangle$ with $\eta = +1$ or belong to $(V \otimes V)_{\mathrm{A}}$ spanned by $\left|\psi\right\rangle \otimes \left|\phi\right\rangle - \left|\phi\right\rangle \otimes \left|\psi\right\rangle$.

Similarly for multiparticle states with $N \geq 2$ we can define $U_{(a,b)}$ being a transposition $\{\mathbf{x}_a, r_a\} \to \{\mathbf{x}_b, r_b\}$ by simlarl action on basis states. Then

$$U_{(a,b)}\left|\Phi\right\rangle = \eta_{(a,b)}\left|\Psi\right\rangle$$

with $\eta_{(a,b)} = \pm 1$ since $U_{(a,b)}$ is a transposition. Now for any permutation π of $\{1, 2, \ldots, N\}$ define U_π to be the operator which applies the permutation π to the basis states like we did with $U_{(a,b)}$. On a general state we have that

$$U_\pi\left|\Psi\right\rangle = \eta_\pi\left|\Psi\right\rangle$$

for some $\eta_\pi = \pm 1$. But the algebra of transpositions implies that

$$\eta_{(a,b)} = \pm 1$$

with the same value for all pairs (a, b). Since π can be obtained as a product of transpositions from IA Groups we have that

$$\eta_\pi = \begin{cases} 1 \\ \operatorname{sgn}(\pi) = (-1)^{\#\text{ of swaps needed for } \pi} \end{cases}$$

with the same alternative for all π.

Hence there are two fundamentally different kinds of particles:

- *Bosons* which obey *Bose-Einstein* statistics where an interchange of identical bosons leave the state unchanged, $\eta = +1$ with multiparticle states in $(V \otimes \cdots \otimes V)_{\mathrm{S}}$
- *Fermions* which obey *Fermi-Dirac* statistics where the interchange of identical fermiions changes the state by a sign, $\eta = -1$ with mutliparticle states in $(V \otimes \cdots \otimes V)_{\mathrm{A}}$

This fundamental principle states that integral spin particles are bosons and half integral spin particles are fermions.

This is confirmed by experiment and can be derived in relativistic quantum mechanics or quantum field theory where it becomes the spin-statistic theorem. The most common elementary particles are fermions such as electrons, protons, neutrons, neutrinos which are all spin $\frac{1}{2}$. So too are quarks, muon, etc.

Particles associated with forces are bosons such as photons, $W^{\pm}$, Z bosons, and gluons which are all spin 1.

4.3.2 Two Particle Examples

Now we know that states of two identical particles belong to

$$(V \otimes V)_{\mathrm{S}} \quad \text{or} \quad (V \otimes V)_{\mathrm{A}}$$

where $V = V_{\text{space}} \otimes V_{\text{spin}}$. It is convenient to construct these by first taking

$$(V_{\text{space}} \otimes V_{\text{space}})_{\mathrm{S\ or A}} \quad \text{and} \quad (V_{\text{spin}} \otimes V_{\text{spin}})_{\mathrm{S\ or A}}$$

separately, then combining to get the desired symmetry overall.

For instance, consider the single particle Hamiltonian $H(\hat{\mathbf{x}}, \hat{\mathbf{p}})$ independent of spin, with non-degenerate energy levels

$$E_0 < E_1 < E_2 < \dots$$

with wavefunctions

$$\psi_0(\mathbf{x}), \psi_1(\mathbf{x}), \dots$$

for states in V_{space}.

So for a pair we can split the Hamiltonians

$$H(\hat{\mathbf{x}_1}, \hat{\mathbf{p}_1}) + H(\hat{\mathbf{x}_2}, \hat{\mathbf{p}_2})$$

and basis of energy eigenfunctions $\psi_i(\mathbf{x}_1), \psi_j(\mathbf{x}_2)$ with energy $E = E_i + E_j$.

The lowest level

$$\Psi_0^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2) = \psi_0(\mathbf{x}_1)\psi_0(\mathbf{x}_2)$$

we have $E = 2E_0$ and in the next level theres no definite symmetry so we do S or A so we get

$$\Psi_1^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}}(\psi_0(\mathbf{x}_1)\psi_1(\mathbf{x}_2) + \psi_1(\mathbf{x}_1)\psi_0(\mathbf{x}_2))$$
$$\Psi_1^{\mathrm{A}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}}(\psi_0(\mathbf{x}_1)\psi_1(\mathbf{x}_2) - \psi_1(\mathbf{x}_1)\psi_0(\mathbf{x}_2)).$$

Now consider spin. For spin 0 V_{spin} is just one state so we can ignore it. But spin-statistics implies that bosons and states are in $(V \otimes V)_{\mathrm{S}}$ or in this case $(V_{\text{space}} \otimes V_{\text{space}})_{\mathrm{S}}$. The lowest level is $\Psi_0^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2)$ with energy $2E_0$ and the next level is $\Psi_1^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2)$ with energy $E_0 + E_1$ both non-degenerate.

With spin $\frac{1}{2}$ we have that V_{spin} has basis $\{|\uparrow\rangle, |\downarrow\rangle\}$ for each particle. So the tensor product has basis $\{|\uparrow\rangle|\uparrow\rangle, |\uparrow\rangle|\downarrow\rangle, |\downarrow\rangle|\uparrow\rangle, |\downarrow\rangle|\downarrow\rangle\}$ or (normalised combinations with definite symmetry

$$\frac{1}{\sqrt{2}}(|\uparrow\rangle|\downarrow\rangle + |\downarrow\rangle|\uparrow\rangle) \qquad \frac{1}{\sqrt{2}}(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle)$$

with the first being part of $(V_{\text{spin}} \otimes V_{\text{spin}})_{\mathrm{S}}$ and the second being part of $(V_{\text{spin}} \otimes V_{\text{spin}})_{\mathrm{A}}$.

Spin-statistics gives that the states must belong to

$$(V \otimes V)_{\mathrm{A}} = (V_{\text{space}} \otimes V_{\text{space}})_{\mathrm{A}} \otimes (V_{\text{spin}} \otimes V_{\text{spin}})_{\mathrm{S}} + (V_{\text{space}} \otimes V_{\text{space}})_{\mathrm{S}} \otimes (V_{\text{spin}} \otimes V_{\text{spin}})_{\mathrm{A}}$$

so at the lowest level we have that

$$\Psi_0^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2)\frac{1}{\sqrt{2}}(|\uparrow\rangle |\downarrow\rangle - |\downarrow\rangle |\uparrow\rangle)$$

and the next level is

$$\begin{gathered}\Psi_1^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2)(1 \text{ state from } (V_{\text{spin}} \otimes V_{\text{spin}})_{\mathrm{A}} \ ) \\ \text{or } \Psi_1^{\mathrm{A}}(\mathbf{x}_1, \mathbf{x}_2)(3 \text{ states from } (V_{\text{spin}} \otimes V_{\text{spin}})_{\mathrm{S}} \ )\end{gathered}$$

with 4 states with energy $E_0 + E_1$.

4.4 Pauli Exclusion Principle and Atomic Structure

A state of N identical fermions can be specified by taking N *distinct* one particle states (each in V) and antisymmetrising to ensure Fermi statistics (producing states in $(V \otimes \cdots \otimes V)_{\mathrm{A}}$). The condition that the one particle states must be distinct (else we get zero) is the *Pauli Exclusion Principle*.

The original application of this was to atomic structure. Consider N electrons bound to a nucleus with Z protons. Ignoring electron-electron interactions the Hamiltonian is

$$H(\hat{\mathbf{x}_1}, \hat{\mathbf{p}_1}) + \cdots + H(\hat{\mathbf{x}_N}, \hat{\mathbf{p}_N})$$

with single particle Hamiltonian

$$H(\hat{\mathbf{x}}, \hat{\mathbf{p}}) = \frac{\hat{\mathbf{p}}^2}{2m} - \frac{Ze^2}{4\pi\varepsilon_0|\hat{\mathbf{x}}|}.$$

For a single electron with states in V_{space} we have joint eigenstates $|n, \ell, m\rangle$ of $H, \mathbf{L}^2, L_3$ with eigenvalues $E_n, \hbar^2\ell(\ell+1), \hbar m$ respetively. Where

$$E_n = -\frac{m}{2}\left(\frac{2e^2}{4\pi\varepsilon_0\hbar}\right)^2 \frac{1}{n^2}$$

and $n = 1, 2, \ldots$ is the principal quantum number and $\ell = 0, 1, \ldots, n-1$ and $m = 0, \pm 1, \ldots, \pm\ell$ gives $2\ell + 1$ states.

Hence the degeneracy level of E_n is

$$1 + 3 + \cdots + (2n-1) = n^2.$$

Now if we include V_{spin} with basis $\{|\uparrow\rangle, |\downarrow\rangle\}$ each electron has $2n^2$ states at energy E_n.

Let's see the strucutre of electrically neutral atoms with $N = 2$. We fill up energy levels, starting with the lowest using Pauli exclusion principle. This gives a rough picture of the periodic table with some qualitative insights into chemical properties. For example we expect unreactive elements for completely fill energy levels (noble gases). For example E_1 filled for $Z = 2$ gives helium and E_1 and E_2 filled for $Z = 10$ gives neon.

5 Perturbation Theory

Few quantum mechanical systems can be solved exactly. In perturbation theory we start from a known system

$$H\left|n\right\rangle = E_n\left|n\right\rangle$$

with $\{|n\rangle\}$ being an orthonormal basis of energy eigenstates. We calculate energies and eigenstates for a *perturbed* Hamiltonian

$$(H + \lambda V)\left|\psi\right\rangle = E\left|\psi\right\rangle.$$

order-by-order in parameter λ with V some Hermitian operator. We assume that E and $\ker\psi$ depend smoothly on λ.

5.1 Non-degenerate Case

Suppose that $\lambda \to 0$ so

$$\begin{aligned}\left|\psi\right\rangle &\to \left|r\right\rangle \\ E &\to E_r.\end{aligned}$$

Use $\{|n\rangle\}$ as a basis for $\lambda \neq 0$ so we can write

$$\psi = \alpha\left|r\right\rangle + \sum_{j\neq r}\beta_j\left|j\right\rangle = \alpha\left(\left|r\right\rangle + \sum_{j\neq r}\gamma_j\left|j\right\rangle\right)$$

where $\alpha, \beta, \gamma_j = \beta_j/\alpha$ are all power series in λ such that with $\alpha \to 1$ and $\beta_j, \gamma_j \to 0$ as $\lambda \to 0$. The aim to calculate coefficients in expansions

$$\begin{aligned}E &= E_r + E_r'\lambda + E_r''\lambda^2 + \ldots \\ \alpha &= 1 + a\lambda + \ldots \\ \gamma_j &= c_j\lambda + \ldots\end{aligned}$$

but first substitute expressions for $|\psi\rangle$ into the Schrödinger equation, so

$$(E_r + \lambda V)\left|r\right\rangle + \sum_{j\neq r}\gamma_j(E_j + +\lambda V)\left|j\right\rangle = E\left(\left|r\right\rangle + \sum_{j\neq r}\gamma_j\left|j\right\rangle\right)$$

cancelling a factor of α, or

$$(E - E_r)\left|r\right\rangle + \sum_{j\neq r}\gamma_j(E - E_j)\left|j\right\rangle = \lambda V\left|r\right\rangle + \lambda\sum_{j\neq r}\gamma_j V\left|j\right\rangle$$

so taking $\langle r|$ of the above equation we get that

$$E - E_r = \lambda\left\langle r\right|V\left|r\right\rangle + \lambda\sum_{j\neq r}\gamma_j\left\langle r\right|V\left|j\right\rangle.$$

So far this is exact (holds for all orders in λ) but now in terms of the series expansion we find that

$$\lambda E_r' + \lambda^2 E_r'' + \cdots = \lambda\left\langle r\right|V\left|r\right\rangle + \lambda^2\sum_{j\neq r}c_j\left\langle r\right|V\left|j\right\rangle + \ldots$$

so we can see that first-order change is given by

$$E_r' = \langle r| V |r\rangle .$$

To find second-order change we need to know c_j, so multiplying with $\langle i|$ instead of $\langle r|$ we get that

$$\gamma_i(E - E_i) = \lambda \langle i| V |r\rangle + \lambda \sum_{j\neq r} \gamma_j \langle i| V |j\rangle + \dots .$$

Again this result is exact, but to leading order in λ we get that

$$\lambda c_i(E - E_i) = \lambda \langle i| V |r\rangle$$

hence

$$c_i = \frac{\langle i| V |r\rangle}{E_r - E_i}$$

using that fact that $E_i \neq E_r$ since we're looking at the non-degenerate case and

$$E_r'' = \sum_{j\neq r} \frac{|\langle j| V |r\rangle|^2}{E_r - E_j}$$

so in summary

$$E = E_r + \lambda \langle r| V |r\rangle + \lambda^2 \sum_{j\neq r} \frac{|\langle j| V |r\rangle|^2}{E_r - E_j}$$

to $O(\lambda^2)$.

Example. We'll return to the harmonic oscillator.

$$H = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = \hbar w \left(a^\dagger a + \frac{1}{2}\right)$$

with states $|n\rangle$, $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$. Let

$$\begin{aligned} V = m\omega^2\hat{x}^2 &= m\omega^2 \frac{\hbar}{2m\omega}(a^2 + {a^\dagger}^2 + aa^\dagger + a^\dagger a) \\ &= \hbar\omega(a^2 + {a^\dagger}^2 + 2a^\dagger a + 1). \end{aligned}$$

Note that

$$\begin{aligned} \langle n| V |n\rangle &= \frac{1}{2}\hbar\omega(2n+1) \\ \langle n+2| V |n\rangle &= \frac{1}{2}\hbar\omega\sqrt{n+1}\sqrt{n+2} \\ \langle n-2| V |n\rangle &= \frac{1}{2}\hbar\omega\sqrt{n}\sqrt{n-1} \;\; \langle m| V |n\rangle = 0 \end{aligned}$$

for all other m. The perturbed energy of the nth level to $O(\lambda^2)$ is

$$\begin{aligned} &E_n + \lambda \langle n| V |n\rangle + \lambda \sum_{m\neq n} \frac{|\langle m| V |n\rangle|^2}{E_n - m} \\ &= \hbar\omega\left(n + \frac{1}{2}\right) + \lambda\hbar\omega\left(n + \frac{1}{2}\right) + \lambda^2\left(\frac{\hbar\omega}{2}\right)^2 \left(\frac{(n+1)(n+2)}{-2\hbar\omega} + \frac{n(n-1)}{2\hbar\omega}\right) \\ &= \hbar\omega\left(n + \frac{1}{2}\right)\left(1 + \lambda - \frac{\lambda^2}{2}\right) \end{aligned}$$

to $O(\lambda^2)$. But this problem can be solved exactly. $H \to H + \lambda V$ is equivalent to $\omega^2 \to \omega'^2 = \omega^2(1+2\lambda)$. New energies are exactly

$$\begin{aligned}\hbar\omega'\left(n+\frac{1}{2}\right) &= \hbar\omega\left(n+\frac{1}{2}\right)(1+2\lambda)^{1/2} \\ &= \hbar\omega\left(n+\frac{1}{2}\right)\left(1+\lambda-\frac{1}{2}\lambda^2+O(\lambda^2)\right).\end{aligned}$$

In general we expect perturbation theory to be valid and useful when $|\lambda|$ is sufficiently small. But we really have an expansion in $\lambda^2 \frac{|\langle i|v|j\rangle|^2}{|E_i - E_j|}$ and similarly quantities. The condition for accuracy is therefore that

$$|\text{energy shifts}| \ll |\text{original energy differences}|$$

so that H eigenvalues are close to $H + \lambda V$ eigenvalues.

In general perturbation series may not converge but are still useful as an asymptotic series. In the harmonic oscillator example we consider the total potential $\frac{1}{2}m\omega^2(1+2\lambda)\hat{x}^2$. For $\lambda < -\frac{1}{2}$ we expect the theory to not make sense since the sign is flipped, which suggests perturbation theory converges for $|\lambda| < \frac{1}{2}$.

Example. (Ground state energy for Helium) For the unperturbed problem, we have two non-interacting electrons of charge $-e$ oribiting a nucleus of charge $+2e$ with Hamiltonian

$$H(\hat{\mathbf{x}}_1, \hat{\mathbf{p}_1}) + H(\hat{\mathbf{x}}_2, \hat{\mathbf{p}_2})$$

with

$$H(\hat{\mathbf{x}}, \hat{\mathbf{p}}) = \frac{\hat{\mathbf{p}}^2}{2m} - \frac{2e^2}{4\pi\varepsilon_0|\hat{\mathbf{x}}|}.$$

One electron state and energies are

$$|n,\ell,m\rangle \quad E_n = -\frac{1}{2}m\left(\frac{2e^2}{4\pi\varepsilon_0\hbar}\right)^2\frac{1}{n^2}.$$

The lowest energy eigenstates for two electrons is

$$|\Psi\rangle = |1,0,0\rangle \otimes |1,0,0\rangle \otimes |\chi\rangle$$

with

$$|\chi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle)$$

spin state. So the two electron wavefunction is

$$\Psi(\mathbf{x}_1, \mathbf{x}_2) + \psi_{1,0,0}(\mathbf{x}_1)\psi_{1,0,0}(\mathbf{x}_2)$$

and

$$\psi_{1,0,0}(\mathbf{x}) = \frac{1}{\sqrt{\pi}}\left(\frac{2}{a_2}\right)^{3/2} e^{-|\mathbf{x}|/a}$$

where

$$a = \frac{1}{2}\frac{4\pi\varepsilon_0}{e^2}\frac{\hbar^2}{m}$$

which is half the Bohr radius. And

$$2E_1 = -m\left(\frac{2e^2}{4\pi\varepsilon_0\hbar}\right)^2.$$

Experimentally, the ground state energy for Helium is different, but we neglected the electron-electron interaction. Treat this as a perturbation so

$$\lambda = \frac{e^2}{4\pi\varepsilon_0} \quad \text{and} \quad V(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{|\mathbf{x}_1 - \mathbf{x}_2|}.$$

The first order correctoin to the ground state energy is

$$\begin{aligned}\lambda \langle\Psi| V |\Psi\rangle &= \lambda \int \mathrm{d}^3\mathbf{x}_1 \int \mathrm{d}\mathbf{x}_2 \Psi(\mathbf{x}_1, \mathbf{x}_2)^* V(\mathbf{x}_1, \mathbf{x}_2)\Psi(\mathbf{x}_1, \mathbf{x}_2) \\ &= \lambda \int \mathrm{d}^3\mathbf{x}_1 \int \mathrm{d}^3\mathbf{x}_2 |\psi_{1,0,0}(\mathbf{x}_1)|^2 |\psi_{1,0,0}(\mathbf{x}_2)|^2 \frac{1}{|\mathbf{x}_1 - \mathbf{x}_2|}\end{aligned}$$

which on evaulation gives us a correction term which matches the experimental value.

5.2 Degenerate Case

Consider the pertubration system

$$(H + \lambda V) |\psi\rangle = E |\psi\rangle$$

and suppose as before that $\lambda \to 0$, but now $E \to E_r = E_s = \ldots$ which are eigenvalues of denegerate states $|r\rangle, |s\rangle, \ldots$. So $|\psi\rangle$ can be written as a linear combination

$$|\psi\rangle \longrightarrow \sum_r a_r |r\rangle .$$

We'll use j to label these other basis states only (the ones with eigenvalues not equal to E_r. For a general λ we'll expand

$$|\psi\rangle = \sum_r \alpha_r |r\rangle + \sum_j \beta_j |j\rangle$$

so we must have that $\alpha_r = a_r + O(\lambda)$ and $\beta_r = O(\lambda)$. Putting $|\psi\rangle$ into the Schrödinger equation, we have that

$$\sum_r \alpha_r (E_r + \lambda V) |r\rangle + \sum_j \beta_j (E_j + \lambda V) |j\rangle = E\left(\sum_r \alpha_r |r\rangle + \sum_j \beta_j |j\rangle\right)$$

Rearraging we get that

$$\sum_r \alpha_r (E - E_r) |r\rangle + \sum_j \beta_j (E - E_j) |j\rangle = \lambda \sum_r \alpha_r V |r\rangle + \lambda \sum_j \beta_j V |j\rangle .$$

We seek $E = E_r + \lambda E' + O(\lambda^2)$ and note that the correction E' is no longer associated with a specific basis state in our chosen set. Take the inner product of the rearranged equation with $\langle s|$ to get that

$$(E - E_s)\alpha_s = \lambda \sum_r \langle s| V |r\rangle + \lambda \sum_j \beta_j \langle s| V |j\rangle .$$

Then to leading order in λ we have that

$$\sum_r \langle s| V |r\rangle a_r = E' a_s.$$

Thus E' is an eigenvalues of the matrix $\langle s| V |r\rangle$ and a_r components of the eigenvector. If there are N degenerate states then we have N solutions for E' in general and the corresponding eigenstates are $\sum_r a_r |r\rangle$.

In the general picture the degeneracy of H is a reflection of some symmetry. The perturbed hamiltonian $H + \lambda V$ generally has less symmetry so levels split apart.

Example. (Particle in a box in 2D) Suppose we have a box $0 \leq x, y \leq a$ where the unperturbed problem has states $|p\ q\rangle$ and energies $E_{pq} = \frac{\hbar^2\pi^2}{2ma^2}(p^2 + a^2)$. The wave functions are

$$\psi_{pq} = \frac{2}{a} \sin\frac{p\pi x}{a} \sin\frac{q\pi y}{a}$$

where $p, q \in \mathbb{N}$. The lowest energy level $E_{11} = \frac{\hbar\pi^2}{ma^2}$ is non-degnerate and the next level

$$E_{12} = E_{21} = \frac{5\hbar^2\pi^2}{2ma^2}$$

with degeneracy 2. Perturbation $V(x, y) = \frac{xy}{a^2}$ which is dimensionless so λ has the dimension of energy. The shift in the lowest energy level is given by

$$\langle 1\ 1| V |1\ 1\rangle = \int_0^a \mathrm{d}x \int_0^a \mathrm{d}y\ \frac{xy}{a^2} |\psi_{11}(x, y)|^2 = \frac{1}{4}.$$

The shifts in degenerate levels given by the eigenvalues of

$$\begin{pmatrix} \langle 1\ 2| V |1\ 2\rangle & \langle 1\ 2| V |2\ 1\rangle \\ \langle 2\ 1| V |1\ 2\rangle & \langle 2\ 1| V |2\ 1\rangle \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$$

which are just $\alpha \pm \beta$ with eigenvalues $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$.

We have that

$$\begin{aligned} \alpha &= \int_0 a\mathrm{d}x \int_0^a \mathrm{d}y\ \frac{xy}{a^2} |\psi_{12}(x, y)|^2 \\ &= \left(\frac{2}{a}\right)^2 \frac{1}{a^2} \left(\int_0^a \mathrm{d}x\ \sin^2\frac{\pi x}{a}\right) \left(\int_0^a \mathrm{d}y \sin^2\frac{2\pi y}{a}\right) \\ &= \frac{1}{4} \end{aligned}$$

and

$$\begin{aligned} \beta &= \int_0^a \int_0^a \mathrm{d}y\ \frac{xy}{a^2} \psi_{12}(x, y)\psi_{21}(x, y) \\ &= \left(\frac{2}{a}\right)^2 \frac{1}{a^2} \left(\int_0^a \mathrm{d}x\ x \sin\frac{\pi x}{a} \sin\frac{2\pi x}{a}\right)^2 \\ &= \left(\frac{4}{3\pi}\right)^4. \end{aligned}$$

So summing up the new ground state energy is

$$\frac{\hbar^2\pi^2}{ma^2} + \frac{\lambda}{4}$$

and the next two levels have energies

$$\frac{5\hbar^2\pi^2}{2ma^2} + \frac{\lambda}{4} \pm \lambda\left(\frac{4}{3\pi}\right)^4$$

and the corresponding states are

$$\frac{1}{\sqrt{2}}\left(|1\ 2\rangle \pm |2\ 1\rangle\right).$$

We expect perturbation theory to be accurate for $\lambda \ll \frac{\hbar^2\pi^2}{ma^2}$.

6 Angular Momentum

6.1 Recap of Orbital Angular Momentum

Consider a particle with position $\hat{\mathbf{x}}$ and momentum $\hat{\mathbf{p}}$. We have that

$$\mathbf{L} = \hat{\mathbf{x}} \times \hat{\mathbf{p}} \quad \text{or} \quad L_i = \varepsilon_{ijk}\hat{x}_j\hat{p}_k.$$

From the relation $[L_i, L_j] = i\hbar\varepsilon_{ijk}L_k$ and $\mathbf{L}^2 = L_iL_i$ which gives that $[\mathbf{L}^2, L_i] = 0$. Note that the commutator above can be derived from the relations

$$[L_i, \hat{x}_j] = i\hbar\varepsilon_{ijk}\hat{x}_k \quad \text{and} \quad [L_i, \hat{p}_j] = i\hbar\varepsilon_{ijk}\hat{p}_k.$$

Consider the action of all operators on wavefunctions. In spherical polar coordinates (r, θ, φ), the L_i involves only angular derivatives and

$$\nabla^2 = \frac{1}{r}\frac{\partial^2}{\partial r^2}(r) - \frac{1}{r^2\hbar^2}\mathbf{L}^2.$$

We have joint eigenstates of $\mathbf{L}^2$ and L_3 which are the spherical harmoincs $Y_{\ell m}(\theta, \varphi)$

$$\mathbf{L}^2 Y_{\ell m} = \hbar^2\ell(\ell+1)Y_{\ell m}$$
$$L_3 Y_{\ell m} = \hbar m Y_{\ell m}$$

for $Y_{\ell m}$ in terms of Legendre polynomials. We have well-behaved solutions of the eigenvalue problem only for

$$\ell = 0, 1, 2, \ldots \qquad m = 0, \pm1, \pm2, \ldots, \pm\ell.$$

If the Schrödinger equation has spherically symmetric potential, then we can seek separable solutions

$$\psi_{\ell m}(\mathbf{x}) = R_\ell(r)Y_{\ell m}(\theta, \varphi)$$

where

$$-\frac{\hbar^2}{2m}\frac{1}{r}\frac{\partial^2}{\partial r^2}(rR_\ell) + \left(\frac{\hbar^2}{2mr^2}\ell(\ell+1) + V(r)\right)R_\ell = ER_\ell$$

from the time-independent Schrödinger equation.

Of particular importance, the behaviour of such solutions under the map $\mathbf{x} \to -\mathbf{x}$ can give us a *parity* of a solution. This is equivalent to

$$r \to r, \quad \theta \to \pi - \theta, \quad \varphi \to \varphi + \pi$$

and it turns out this gives that

$$Y_{\ell m}(\pi - \theta, \varphi + \pi) = (-1)^\ell Y_{\ell m}(\theta, \varphi).$$

Hence

$$\psi_{\ell m}(-\mathbf{x}) = (-1)^\ell \psi_{\ell m}(\mathbf{x}).$$

Example. There is an important case where we have a system of two particles interacting through a potential which depends only on their relative separation.

$$\begin{aligned} H &= \frac{\hat{\mathbf{p}_1}^2}{2m_1} + \frac{\hat{\mathbf{p}_2}^2}{2m_2} + V(|\hat{\mathbf{x}_1} - \hat{\mathbf{x}_2}|) \\ &= \frac{\hat{\mathbf{P}}^2}{2m} + \frac{\hat{\mathbf{p}}^2}{2m} + V(|\hat{\mathbf{x}}|) \end{aligned}$$

where

$$\begin{aligned} \hat{\mathbf{X}} &= \frac{m_1\hat{\mathbf{x}_1} + m_2\hat{x_2}}{M}, \qquad \hat{\mathbf{x}} = \hat{\mathbf{x}_1} + \hat{\mathbf{x}_2} \\ \hat{\mathbf{P}} &= \hat{\mathbf{p}_1} + \hat{\mathbf{p}_2}, \qquad \hat{\mathbf{p}} = \frac{m_2\hat{\mathbf{p}_1} - m_1\hat{\mathbf{p}_2}}{N} \end{aligned}$$

where $M = m_1 + m_2$ and $m = \frac{m_1 m_2}{M}$ is the reduced mass.

The centre of mass dynamics is trivial if no external force/potential and relative motion is governed by spherically symmetric potential. The wavefuntcion is

$$\Psi(\mathbf{X}, \mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{X}} \psi_{\ell m}(\mathbf{x}).$$

Note that we have that under $\mathbf{x} \to -\mathbf{x}$, $\Psi \to (-1)^\ell \Psi$ we have an interchange of particles.

6.2 General Analysis of Angular Momentum Eigenstates

Consider

$$\mathbf{J} = (J_1, J_2, J_3)$$

which are such that $J_i^\dagger = J_i$ and

$$[J_i, J_j] = i\hbar\varepsilon_{ijk} J_k.$$

Define $\mathbf{J}^2 = J_i J_i$ and it follows that $\mathbf{J}^2$ is also Hermitian and we get that

$$[\mathbf{J}^2, J_i] = 0.$$

Seek all possible joint eigenstates of $\mathbf{J}^2$ and J_3. First observe that if we have $\mathbf{J}^2 |\psi\rangle = \lambda |\psi\rangle$ then we get that

$$\lambda \langle\psi| \mathbf{J}^2 |\psi\rangle = \sum_{i=1}^{3} ||J_i |\psi\rangle||^2 \geq 0$$

so it's convenient to set $\lambda = \hbar j(j+1)$ where we can take $j \geq 0$ since λ is unchanged if $j \to -(j+1)$. Now label states $|j\ m\rangle$ where

$$\begin{aligned}\mathbf{J}^2 |j\ m\rangle &= \hbar^2 j(j+1) |j\ m\rangle \\ J_3 |j\ m\rangle &= \hbar m |j\ m\rangle .\end{aligned}$$

the analyse allowed values of j, m define

$$J_\pm = J_1 \pm J_2, \quad (J_\pm)^\dagger = J_\mp$$

and work with these. It's easy to check that

$$\begin{aligned}[J_3, J_\pm] &= \pm\hbar J_\pm \\ [J_+, J_-] &= 2\hbar J_3 \\ [\mathbf{J}^2, J_\pm] &= 0\end{aligned}$$

and

$$\begin{aligned}J_+ J_- &= \mathbf{J}^2 - J_3^2 + \hbar J_3 \\ J_- J_+ &= \mathbf{J}^2 - J_3^2 - \hbar J_3.\end{aligned}$$

Now given any state $|j\ m\rangle$ we notice that $J_\pm |j\ m\rangle$ gives an eigenstate of $\mathbf{J}^2$ with j unchanged and J_3 with $m \to m \pm 1$, provided the new states are non-zero.

$$\begin{aligned}J_3(J_\pm |j\ m\rangle &= ([J_3, J_\pm] + J_\pm J_3) |j\ m\rangle \\ &= (\pm\hbar J_\pm + J_\pm \hbar m) |j\ m\rangle \\ &= \hbar(m \pm 1) J_\pm |j\ m\rangle .\end{aligned}$$

We can compare with the oscillator where $[N, a^\dagger] = a^\dagger$ and $[N, a] = -a$ which implies that $a^\dagger |n\rangle$ and $a |n\rangle$ have eigenvalues $n \pm 1$ under N.

To find out whether or not the new states vanish, we compute

$$\begin{aligned}||J_+ |j\ m\rangle ||^2 &= \langle j\ m| J_- J_+ |j\ m\rangle \\ &= \langle j\ m| \mathbf{J}^2 - J_3^2 - \hbar J_3 |j\ m\rangle \\ &= \hbar^2(j(j+1) - m(m+1)) \\ &= \hbar^2(j-m)(j+m+1) \geq 0\end{aligned}$$

and similarly

$$\begin{aligned}||J_- |j\ m\rangle ||^2 &= \langle j\ m| J_+ J_- |j\ m\rangle \\ &= \langle j\ m| \mathbf{J}^2 - J_3^2 + \hbar H_3 |j\ m\rangle \\ &= \hbar^2(j(j+1) - m(m-1)) \\ &= \hbar^2(j+m)(j-m+1) \geq 0.\end{aligned}$$

The inequalities follow because the inner product is positive definite and we have that

$$j \geq m \geq -(j+1) \qquad j+1 \geq m \geq -j$$

hence

$$j \geq m \geq -j.$$

In addition, consider starting from some normalised state $|j\ m\rangle$ if the states $J_+^n |j\ m\rangle$ are non-vanishing then they have J_3 eigenvalues $\hbar(m+n)$, but these are subject to bound $j \geq m_n$; also to avoid a contradiction, there must be an integer k with $j = m + k$ and hence $J_+^n |j\ m\rangle = 0$ for all $n > k$. Similarly if $\mathbf{J}^n |j\ m\rangle$ are non-vanishing then they have J_3 eigenvalues $\hbar(m-n)$ and $m-n \geq -j$. There mus tbe an integer k' such that $-j = m-k'$ so that $J_-^n |j\ m\rangle = 0$ for all $k' > 0$. Thus $j = m+k$ and $j = -m+k'$ for $k.k'$ integers. And for a given value of j we have states $|j\ m\rangle$ with $m = -j, -j+1, \ldots, j$ for a total of $2j+1$ states (for fixed j). Our general analysis gives two possibilities. That j is integral so $m = 0, \ldots, \pm j$ or j is half-integral so $m = \pm\frac{1}{2}, \frac{3}{2} \ldots, \pm j$ with an odd/even number of states respectively. These cases correspond to $2j = k + k'$ even or odd in the analysis above. The first possibility is realised in orbital angular momentum and the states $|j\ m\rangle$ correspodn to wavefunctions $\psi_{jm}(\mathbf{x})$ or $Y_{jm}(\theta, \varphi)$ discussed in the previous part. The second possibility cannot arise in this way: there are no solutions of differential equations which are well-behaved when j, m are half-integral. Such states must correspond to intrinstic angular momentum or spin and this can be half-integral or integral.

from analysis above, we can choose are joint eigenstates to be related by

$$\begin{aligned} J_+ |j\ m\rangle &= \hbar[(j-m)(j+m+1)]^{1/2} |j\ m+1\rangle \\ J_- |j\ m\rangle &= \hbar[(j+m)(j-m+1)]^{1/2} |j\ m-1\rangle . \end{aligned}$$

All states can be generated from any given state, but often it is convenient to start from the *top state* which is corresponds to the maximum eigenvalues of J_3 so $J_+ - |j\ j\rangle = 0$. All others are therefore obtained by $(J_-)^n |j\ j\rangle$ up to normalisation.